

can enable engineers to design and evaluate products requiring centimetre-level positioning capabilities. The evaluation kit features an open micro-controller along with a high-precision GNSS (HPG) positioning with real-time kinematic, dead reckoning, cellular, Wi-Fi, and Bluetooth communications, and the necessary antennas. The u-Blox kit is helpful for a wide variety of applications and can assist engineers working in areas such as micromobility and low-speed robotics.

u-Blox

<https://www.u-blox.com>

Multi-band GNSS module

Taoglas TFM.110A is a high-precision, multi-band GNSS front ends module that can be used for various applications, including precision agriculture, automotive, robotics, and unmanned aerial vehicles. It offers ease of integration as it comes in a single-package solution that combines impedance matching, careful filter selection, and low-noise design for easy, drop-in use with any antenna or GNSS receiver. The module supports L1, L2, and L5 bands; the two-stage amplifier solution drives high gain and low power consumption. The module offers low noise figures across the full signal chain for signal accuracy, integrity, and continuity.

Taoglas

<https://www.taoglas.com>

DC/DC converter for BMS

Flex Power's BMR4912510/871 is an adjustable quarter-brick DC/DC converter for BMS applications. The converter module delivers a nominal output of 24V with 40-60V input at a power level of up to 1300W. The output voltage of the DC-DC converter can be adjusted between 16V and 26.4V using its 4-pin PMBus digital interface. This wide adjustment range makes it particularly useful to support battery management solutions



used in remote radio base station installations and high-power applications where conversion from 48V to 24V is necessary.

Flex

<https://flexpowermodules.com>

Surface-mount DC/DC converter

RECOM's RSH3 is a high-power 3W model of DC/DC converter in a surface-mount format with wide input ranges. The DC/DC converter has a nominal input of 12V and 24V with outputs ratings of 5V, 12V, 15V, 24V, $\pm 12V$, and $\pm 15V$. The model is suitable for use in battery-operated systems. The converter offers various safety features, such as short-circuit and overcurrent protection, and by using a simple external filter the converter can meet the EN 55032 class A or B limits for input noise. The RSH3 can find use in various applications, including ground-loop isolation for industrial sensors and bus communication networks.

Recom power

<https://recom-power.com>



SOFTWARE AND TOOLS

Raspberry Pi OS

Raspberry Pi OS now gets major enhancements in text based search, network management, and access to the Pi's camera system. The new OS features a new searchable main menu, which is now searchable by text and provides instant results as you type. The biggest addition to this release is the Network-Manager, which adds a bunch of extra features. It allows you to easily connect to wireless networks with hidden SSIDs and makes managing VPN connections more straightforward. Moreover, it allows you to easily configure a Raspberry Pi as a Wi-Fi access point.

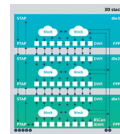
Raspberry Pi Foundation

<https://www.raspberrypi.com>



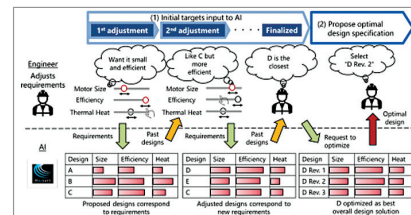
Software for complex ICs

Siemens Digital Industries Software has released a new software solution, called Tessent Multi-die. It can help simplify critical design-for-test tasks for complex integrated circuits based on 2.5D and 3D architectures. The new software can seamlessly work with other software tools using an integrated Tessent platform, thus reducing test time up to 4X. It enables designers to rapidly generate IEEE 1838-compliant hardware featuring 2.5D and 3D IC architectures. Siemens Digital Industries Software <https://resources.sw.siemens.com>



AI tool to design motors

Mitsubishi Electric Corporation and Toshiba Mitsubishi-Electric Industrial Systems Corporation (TMEIC) have jointly developed an electrical motor-design support system. It uses Mitsubishi Electric's Maisart AI technology



to design electrical motors for pumps, compressors, and fans. The design system is able to achieve or even surpass the performance level of conventional design methods deployed manually by skilled engineers. The AI based motor-design system is able to produce ideal design specifications that satisfy all performance targets covering every category, such as motor size, power efficiency, and heat generation, thus reducing the time required for validation and creating new iterations of the design. This AI tool can also help junior engineers to produce motor designs of the same quality as those produced by senior engineers manually.

Mitsubishi Electric Corporation

<https://www.mitsubishielectric.com>